



Hands-on Lab: AI Product Development Challenges and Opportunities

Estimated duration: 30 minutes

Introduction

We illustrated the four stages of the AI Product Development Process. Ideation or Innovation, Data Management, Research & Development, and Deployment. We also provided an overview of how this process integrates with the Product Management Lifecycle and a list of activities required to implement an AI product development project successfully.

Objectives:

In this lab, you will perform the following tasks:

- Analyze a Smart Virtual Assistant development scenario
- Act as an AI Product Manager
 - Map required activities to the AI Product Development Process
 - Define key roles and responsibilities required to plan and implement the required product successfully
 - Identify challenges and risks you may need to overcome and provide mitigation responses
 - Validate your finding using ChatGPT

Prerequisites:

- Ensure that you have viewed the following:
 - Video: Stages of AI Product Development
 - Video: AI Integration in Product Management Lifecycle
 - Reading: AI Support for Product Management Success
- You must have access to Microsoft Word, either through the desktop or online version. If you want to sign up for Microsoft Office 365 and need

help, view the [Sign up for Microsoft 365 lab](#).

Task 1: Review the following Smart Virtual Assistant product development scenario

Your firm is geographically spread out over all of North America and Europe. Leadership believes that the development of a proprietary AI-driven Smart Virtual Assistant to serve everyone in the company will improve customer support and efficiency, reduce costs, alleviate stress, and facilitate more data-driven decisions. You have been assigned as the AI Product Manager to plan and develop this product.

The following features and functions will be required in the product:

- Management expects high user engagement with the system. They want a 100% adoption rate within 3 months of product launch. In addition, they want at least a 4.5 satisfaction score among all company users.
- They want to identify routine tasks that the system can address, analytics that show how productivity will increase, and resource requirements that will be reduced. They expect a data-based cost-benefit analysis as part of the product concept planning.
- They require that all responses be accurate and based on credible data. They expect a response accuracy rate of 98% or more when the system is prompted. In addition, they expect the system to be bias-free and secure.
- The new system must effectively integrate with the existing corporate architecture and applications. In addition, the system must only be available to authorized users in the firm.
- Management expects the initial capability to support 20,000 employees, which will be scalable to 25,000 at the end of year one.
- The system must comply with all privacy laws, ethical guidelines, and industry standards.

Task 2: Download the Scenario Evaluation Worksheet

Keeping the CTRL key pressed, click to download the [Scenario Evaluation Worksheet](#) that provides tables to complete this assignment.

Note: Use the COMMAND key instead of CTRL on Mac systems.

Task 3: Analyze and document requirements

- Map requirements to the AI Product Development process.
- Include at least three requirements in each of the following stages:
 - Ideation and Innovation
 - Data Management
 - Research & Development
 - Deployment

Task 4: Identify the AI product management team

- Define at least four roles required to plan and develop this proposed AI product.
- List the basic responsibilities associated with each role.
- Don't forget to include the AI product manager!

Task 5: Analyze challenges and risks

- Analyze challenges and risks.
- List at least four challenges or risks and your response to mitigate them.

Task 6: Hands-on practice with ChatGPT

- Use ChatGPT to validate your analysis from Tasks 3, 4, and 5
- Input each prompt separately into ChatGPT:
 - **Prompt 1:** Review the Smart Virtual Assistant scenario and define at least three requirements in each AI product development stage.
 - **Prompt 2:** Define at least four roles required for planning and development, including responsibilities, and the AI Product Manager.
 - **Prompt 3:** Analyze challenges and risks, listing at least four with mitigation responses.

Task 7: Review the solution

Keeping the CTRL key pressed, click to download the [Product Concept Potential Solutions](#) document.

Note: Use the COMMAND key instead of CTRL on Mac systems.

- Compare your analysis to the recommended solution.
- Update your analysis as required.

Note: Your solution does not need to match this solution. You may identify items not included in the recommended solution. That is fine.

Task 8: Compare and reflect

- Compare your initial analysis, ChatGPT output, and Potential Solution.
- Reflect on similarities, differences, and insights gained.
- Update your analysis and understanding accordingly.

Task 9: Save your final analysis

- Save your final analysis as a Word or PDF file for future reference.